The second law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain the direction of heat transfer and the limits of energy conversion.

SUCCESS CRITERIA - I can...

1. Predict spontaneous net heat flow.
2. Explain why moving heat from cold to hot needs an additional process.
3. Distinguish energy conservation from process direction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “spontaneous” and “heat” to explain your model or result. Sentence starters are optional.

spontaneous: able to occur without the specified external work input

heat: energy transferred because of a temperature difference

work: energy transferred by a force along a displacement

01 Bring back an earlier idea

A 0.3 kg metal cools from 91 to 31°C while 0.25 kg water warms from 11 to 31°C.
Use water c=4200; neglect cup/losses.
Find metal c.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $Q_{hot} = Q_{cold} + W_{in}$.
Refrigerator cycle: heat amounts are positive magnitudes. Work input moves heat from cold to hot; spontaneous flow is hot to cold.

Worked reasoning: Hot and cold blocks in contact exchange energy until thermal equilibrium. Total energy can be conserved while the temperature difference and ability to drive a heat engine decrease.

Your turn: Draw before-and-after temperature states and the spontaneous heat-flow direction.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

A 50°C block touches a 10°C block.
Draw net heat flow and predict their trend toward equilibrium.

Sentence starter: I predict ... . My model supports this because ... .



50°C 10°C

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Solve and justify

A refrigerator removes 200 J from the cold reservoir using 50 J of work.
Find heat released and coefficient of performance Q_c/W.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A proposed machine takes heat from one reservoir and turns all of it into work in a repeating cycle. What thermodynamic limitation is missing?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 0.4 kg object moving at 6 m/s stops through friction over 0.8 m.
Find energy transformed and average opposing-force magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Two bodies exchange heat in an isolated system and reach the same temperature. Is the total energy lost? What happens to the ability to extract work from their temperature difference?

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A refrigerator removes 200 J from a cold space while receiving 80 J electrical work.
Find heat delivered to the room and explain why it is larger than 200 J.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3